# Personal Statement - MSc in Artificial Intelligence, City University of Hong Kong (Dongguan)

On the multi-robot platform I develop for at Kaihong, a high-level task plan cannot be sent directly to a robot. It must pass through an action-proposal, policy-check, execution and state-feedback chain before any actuator is allowed to move. Building that chain has been the most instructive part of my robotics work so far, because it is the point where an AI decision becomes a physical action, and designing it made the gap in my training clear. I can make a robot balance, navigate and follow a trajectory, and I can deploy a vision model to an edge device, but my experience with machine learning has so far been mainly practical; I now need to understand its foundations, failure modes and system-level safeguards. The MSc in Artificial Intelligence is where I intend to close that gap.

I am a final-year Computer Science student at Shenzhen Technology University, ranked 6 of 156, with control theory courses added to the core curriculum. My AI experience began with a personal project, an OpenHarmony smart gardening system, in which I built a YOLOv5 inference chain quantised to INT8 on RK3588, a LangGraph decision chain with a rule-based fast path for extreme states, and a device-side irrigation state machine, and integrated them into a closed loop from camera to pump. Despite its small scale, it exposed questions that become more important in larger systems: how to interpret a detector's confidence, when a learned policy should yield to a rule, and how to keep a physical actuator safe when the decision module is uncertain.

The rest of my experience is in robot control and navigation on real hardware. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society I was responsible for the balance control of a dual-arm ballbot, combining a deployed cascaded PID controller with a linearised model and LQR analysis, and the chassis maintained balance while the arms moved. In laboratory projects I implemented gain-scheduled LQR and virtual model control on a wheel-legged robot and the full localisation, planning and MPC tracking chain on a car-like vehicle. At Future Power I worked on VR teleoperation of a dual-arm robot with constrained inverse kinematics. Together, these projects showed me the limits of fixed pipelines: I tuned parameters by hand, fused states with fixed filters and used planners that assumed a largely static environment. Learning-based perception, state estimation and control are the next direction I want to pursue, but I want to do so with a proper understanding of their mathematical foundations and failure modes.

The programme's stated outcomes match what I need: to apply AI development tools and propose solutions, to design AI systems rather than only call them, and to identify the key issues in a specific AI area and address them. My specific area is embodied intelligence for mobile robots, which draws on machine learning, deep learning for perception, reinforcement learning for control and the systems discipline to run all of it on hardware. The two-year structure gives me room to take the foundational courses first and then devote a focused project to robot learning, ideally on a physical platform.

After the degree I intend to work as a robotics algorithm engineer in the Greater Bay Area, on perception, navigation and learning-based control for mobile robots. Dongguan sits inside the Greater Bay Area's robotics and smart-manufacturing belt, where this kind of work is being put into production. Across several projects I have taken control and decision modules from design to hardware validation. I now want the AI foundation needed to make such systems improve from data and interaction rather than rely only on fixed rules.